

WORCESTER, South Africa

WMO: 688210

Lat: **33.664S**Lon: **19.419E**Elev: **653**StdP: **14.35**Time Zone: **2.00 (E02)**Period: **98-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	35.8	38.2	31.4	26.3	48.3	33.1	28.3	47.7	37.5	55.1	34.4	57.6	2.1	70	0.465

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	22.4	95.3	67.8	92.0	67.2	89.1	66.6	71.2	86.9	69.9	85.1	68.7	83.5	11.3	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
66.6	100.3	77.5	65.1	95.0	76.5	63.4	89.5	75.2	35.4	86.7	34.2	85.2	33.2	83.6	76.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 31.3	27.3	24.1	32.0	102.9	1.1	4.5	31.3	106.2	30.6	108.8	30.0	111.4	29.3	114.7		
(5)			31.0	72.8	1.0	1.9	30.3	74.2	29.8	75.3	29.2	76.4	28.5	77.8	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	65.1	76.4	76.3	72.5	66.7	60.4	54.0	53.2	54.6	59.2	65.4	69.3	74.2	(6)	(7)
	DBStd	9.73	4.62	4.53	5.26	5.70	4.47	4.34	4.62	4.30	4.88	5.94	5.25	5.00	(7)	(8)
	HDD50	37	0	0	0	0	0	12	17	7	1	0	0	0	(8)	(9)
	HDD65	1502	0	1	5	47	155	332	367	323	183	66	21	2	(9)	(10)
	CDD50	5559	819	737	698	500	322	131	117	150	277	478	580	750	(10)	(11)
	CDD65	1548	355	318	238	97	11	1	2	1	9	78	151	287	(11)	(12)
	CDH74	11973	2908	2527	1587	636	101	14	19	20	103	646	1090	2322	(12)	(13)
	CDH80	4678	1271	1089	561	185	11	1	3	2	19	215	352	969	(13)	(14)
(14) Wind	WSAvg	9.3	9.9	9.2	8.8	7.7	8.6	9.9	9.7	9.6	9.5	9.2	9.7	9.8	(14)	(15)
(15) Precipitation	PrecAvg	10.7	0.3	0.3	0.4	1.0	1.2	1.8	1.6	1.5	0.9	0.8	0.4	0.3	(15)	(16)
	PrecMax	18.7	4.0	2.9	2.3	4.1	4.9	5.2	5.3	6.3	4.5	4.3	2.7	1.3	(16)	(17)
	PrecMin	5.4	0.0	0.0	0.0	0.0	0.0	0.1	0.2	0.1	0.0	0.0	0.0	0.0	(17)	(18)
	PrecStd	3.8	0.7	0.6	0.5	1.1	1.1	1.3	1.2	1.3	1.0	0.9	0.6	0.4	(18)	(19)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	99.5	99.1	94.4	91.2	81.6	76.1	76.3	76.7	83.0	94.0	93.2	99.2	(19)	(20)
		MCWB	69.2	68.9	67.8	65.6	59.3	56.9	56.3	57.6	61.4	66.7	65.8	68.8	(20)	(21)
	2%	DB	94.4	94.1	89.7	85.4	76.5	68.1	68.9	69.6	76.9	86.0	88.0	92.9	(21)	(22)
		MCWB	68.3	67.7	66.9	64.7	59.4	55.2	55.4	56.9	59.4	63.1	64.1	67.1	(22)	(23)
	5%	DB	90.5	90.2	86.1	81.1	73.0	64.7	64.9	65.9	72.5	81.0	84.1	89.1	(23)	(24)
		MCWB	67.4	67.5	65.8	63.8	58.8	54.5	54.2	55.1	57.9	61.5	63.2	66.5	(24)	(25)
	10%	DB	87.0	86.7	82.6	77.2	69.6	62.1	61.7	62.8	68.7	76.7	80.4	85.5	(25)	(26)
		MCWB	66.6	66.8	65.0	62.5	57.4	53.5	53.2	53.9	56.3	60.3	62.1	65.6	(26)	(27)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	73.1	72.8	71.2	68.1	63.2	59.0	59.4	60.7	63.4	68.5	69.0	72.1	(27)	(28)
		MCDB	90.5	87.9	85.8	83.5	74.9	68.6	70.4	70.7	76.2	89.0	85.4	89.6	(28)	(29)
	2%	WB	71.1	71.1	69.2	66.5	61.4	57.4	56.8	58.2	61.0	65.3	66.9	70.0	(29)	(30)
		MCDB	87.1	85.8	83.5	80.6	71.7	65.0	64.9	66.6	72.7	81.1	81.4	86.3	(30)	(31)
	5%	WB	69.5	69.7	67.8	65.0	59.9	55.8	55.2	56.4	59.1	63.3	65.2	68.4	(31)	(32)
		MCDB	84.0	84.4	81.7	77.5	69.5	62.8	62.6	64.1	70.7	76.5	79.4	83.8	(32)	(33)
	10%	WB	68.0	68.3	66.4	63.6	58.6	54.5	53.8	54.7	57.2	61.6	63.6	66.9	(33)	(34)
		MCDB	82.8	82.5	79.2	74.8	67.5	60.9	60.8	61.8	67.5	73.6	77.4	81.9	(34)	(35)
(35) Mean Daily Temperature Range	5% DB	MDBR	22.4	22.1	20.1	19.9	19.2	17.6	18.1	17.5	18.6	21.1	21.7	22.3	(35)	(36)
		MCDBR	28.0	27.7	25.0	26.5	25.4	22.5	24.4	23.5	26.1	28.9	27.7	28.1	(36)	(37)
	5% WB	MCWBR	7.5	7.5	6.8	8.9	11.4	12.9	13.8	12.4	10.9	9.9	8.5	7.8	(37)	(38)
		MCWBR	24.8	23.9	22.0	23.1	21.8	18.0	20.1	20.3	23.9	24.8	24.5	25.0	(38)	(39)
(39) Clear-Sky Solar Irradiance	taub		0.341	0.341	0.335	0.317	0.309	0.300	0.303	0.327	0.338	0.353	0.336	0.341	(39)	(40)
		taud	2.499	2.503	2.514	2.564	2.573	2.570	2.552	2.477	2.415	2.396	2.486	2.492	(40)	(41)
	Ebn at Noon		316	310	300	288	274	268	273	280	294	302	316	317	(41)	(42)
		Edh at Noon	36	35	32	27	24	23	24	29	35	38	36	37	(42)	(43)
(43) All-Sky Solar Radiation	RadAvg		2616	2354	1907	1387	991	833	927	1164	1603	2065	2447	2633	(43)	(44)
	RadStd		79	63	72	69	61	54	52	54	73	110	80	96	(44)	(45)

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (15 neighbors)		+0.65	+0.88	N/A	N/A	+0.95	+0.85	-4	-133	+200	+110	

Nomenclature: See separate page